



FINAL YEAR PROJECT
2025 — 2026 · HITU

v1.0 · LIVE

OSC

/ ORCHESTRATION
SECURITY CENTER

An AI-Driven Security Orchestration Platform for SMEs

PROJECT TEAM

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Background & Fundamentals

Bridges the cybersecurity gap for **SMEs** lacking dedicated SOC teams. An **AI-driven DAST** orchestration platform that automates the complete vulnerability assessment lifecycle within a single dashboard.



43%

of cyberattacks target SMEs



1000 → 5

raw logs to action items

IT admins suffer from **tool fragmentation** (Nmap, Nuclei, SIEM) and **alert fatigue**. Enterprise tools remain prohibitively expensive — OSC closes that gap.

02



Architecture & Pipeline

• SYSTEM TOPOLOGY



React 18 Frontend

Vite · Tailwind · WebSockets



FastAPI Backend

Async · Agent Pipeline · REST



Postgres



Redis



Elastic



Wazuh

• THE 4-STAGE AGENT PIPELINE



1 Reconnaissance

Nmap discovers network assets & services



2 Targeted Chaining

Deterministic routing → Nuclei templates



3 Validation

Filters false positives (confidence ≥ 0.6)



4 Risk Scoring

